

MA 5104: Statistical Inference and Learning

Module 1: Statistical Models and Likelihood

Lecture 1: Why Do We Need Statistical Models?

Dr. Rakhi Singh
Department of Mathematics
IIT Madras

Fall 2026

Why Do We Need Statistical Models?

Today's Big Question: Why don't statisticians trust intuition?

Why Do We Need Statistical Models?

Today's Big Question: Why don't statisticians trust intuition?

By the end of today's lecture, you should believe that...

→ Human intuition is often unreliable when reasoning under uncertainty.

Why Do We Need Statistical Models?

Today's Big Question: Why don't statisticians trust intuition?

By the end of today's lecture, you should believe that...

- Human intuition is often unreliable when reasoning under uncertainty.
- Statistical models provide a **principled framework** for understanding uncertain phenomena.

Why Do We Need Statistical Models?

Today's Big Question: Why don't statisticians trust intuition?

By the end of today's lecture, you should believe that...

- Human intuition is often unreliable when reasoning under uncertainty.
- Statistical models provide a **principled framework** for understanding uncertain phenomena.
- Every statistical analysis begins with assumptions about how data are generated.

Why Do We Need Statistical Models?

Today's Big Question: Why don't statisticians trust intuition?

By the end of today's lecture, you should believe that...

- Human intuition is often unreliable when reasoning under uncertainty.
- Statistical models provide a **principled framework** for understanding uncertain phenomena.
- Every statistical analysis begins with assumptions about how data are generated.

A Thought to Keep in Mind

"Statistics is not about numbers. It is about reasoning under uncertainty."

Principled framework: an approach built on explicit assumptions and logically consistent reasoning, producing reproducible conclusions instead of relying on intuition or gut feeling.

Case study

Linda is **31 years old, single, outspoken, and very bright.**

Case study

Linda is **31 years old**, **single**, **outspoken**, and **very bright**.
She majored in **philosophy**.

Case study

Linda is **31 years old, single, outspoken, and very bright.**

She majored in **philosophy.**

As a student, she was deeply concerned with issues of **discrimination** and **social justice**, and also participated in **anti-nuclear demonstrations.**

Case study

Linda is **31 years old**, **single**, **outspoken**, and **very bright**.

She majored in **philosophy**.

As a student, she was deeply concerned with issues of **discrimination** and **social justice**, and also participated in **anti-nuclear demonstrations**.

Reflect on Linda

Case study

Linda is **31 years old, single, outspoken, and very bright.**

She majored in **philosophy.**

As a student, she was deeply concerned with issues of **discrimination** and **social justice**, and also participated in **anti-nuclear demonstrations.**

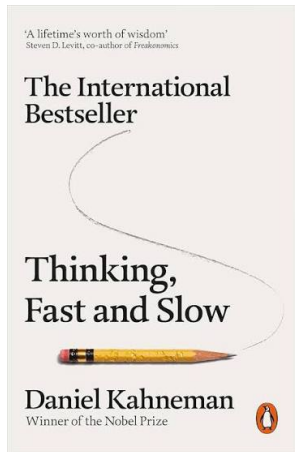
Reflect on Linda

Which of the following is **more probable?**

Option A: Linda is a **bank teller.**

Option B: Linda is a **bank teller** and is **active in the feminist movement.**

Case study



Please read it if you have not!

Linda is **31 years old, single, outspoken, and very bright.**

She majored in **philosophy.**

As a student, she was deeply concerned with issues of **discrimination** and **social justice**, and also participated in **anti-nuclear demonstrations.**

Reflect on Linda

Which of the following is **more probable?**

Option A: Linda is a **bank teller.**

Option B: Linda is a **bank teller** and is **active in the feminist movement.**

Why Did We Get It Wrong?

Your Brain Solved the Wrong Problem

The question was: **“Which event is more probable?”**

Why Did We Get It Wrong?

Your Brain Solved the Wrong Problem

The question was: **“Which event is more probable?”**

Instead, we answered: **“Which description seems more plausible?”**

Why Did We Get It Wrong?

Your Brain Solved the Wrong Problem

The question was: **“Which event is more probable?”**

Instead, we answered: **“Which description seems more plausible?”**

Representativeness Heuristic

We naturally judge events by how well they *resemble* a familiar story or stereotype, rather than by the laws of probability.

Why Did We Get It Wrong?

Your Brain Solved the Wrong Problem

The question was: **“Which event is more probable?”**

Instead, we answered: **“Which description seems more plausible?”**

Representativeness Heuristic

We naturally judge events by how well they *resemble* a familiar story or stereotype, rather than by the laws of probability.

Philosophy; Social justice; Anti-nuclear protests; → “Sounds like Linda.”

Statistical thinking: Ignore whether the story is convincing. Ask only

$$P(A \cap B) \leq P(A)$$

Probability, not plausibility, determines the answer.

Statistics Begins with Uncertainty

Many scientific questions cannot be answered with certainty.

Examples

- Will it rain in Chennai tomorrow?
- Will a patient recover after receiving a treatment?
- Will a customer default on a loan?
- Will this machine fail within the next year?
- Will the next observation exceed 100?

Observation

The outcome is unknown *before* it is observed.
Different outcomes are possible.

Statistics is the science of reasoning under uncertainty.

Deterministic versus Stochastic Systems

Deterministic

Same input
↓
Same output

Examples

- Newton's laws
- Ohm's law
- Matrix multiplication
- Sorting algorithms

Stochastic

Same input
↓
Different possible outcomes

Examples

- Disease progression
- Rainfall
- Financial markets
- Customer behavior

Deterministic versus Stochastic Systems

Deterministic

Same input
↓
Same output

Examples

- Newton's laws
- Ohm's law
- Matrix multiplication
- Sorting algorithms

Stochastic

Same input
↓
Different possible outcomes

Examples

- Disease progression
- Rainfall
- Financial markets
- Customer behavior

Key Insight: Statistics becomes necessary whenever variability is an intrinsic part of the system!

Where Does Variability Come From?

Suppose two patients receive

- the same treatment,
- at the same hospital,
- under the same protocol.

Where Does Variability Come From?

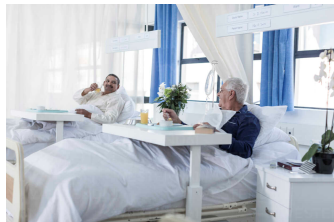
Suppose two patients receive

- the same treatment,
- at the same hospital,
- under the same protocol.

Yet their responses are different.

Possible Reasons:

- Biological variation
- Environmental effects
- Measurement error
- Unknown covariates



Where Does Variability Come From?

Suppose two patients receive

- the same treatment,
- at the same hospital,
- under the same protocol.

Yet their responses are different.

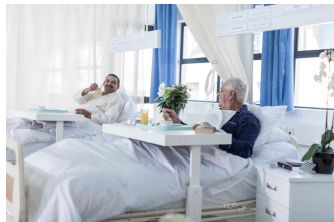
Possible Reasons:

- Biological variation
- Environmental effects
- Measurement error
- Unknown covariates

Observation

Observed variability usually has multiple sources.

Statistics attempts to understand and quantify them.



Signal versus Noise

$$\text{Observed Data} = \text{Signal} + \text{Noise}$$

Signal

The systematic component that reflects the underlying phenomenon.

Examples: Relationship b/w dosage and recovery, Effect of fertilizer on yield

Noise

Variation that is not completely explained by the variables under study. Sources: Meas. error, Environm. fluctuations, Unobserved variables, Random chance

Signal versus Noise

$$\text{Observed Data} = \text{Signal} + \text{Noise}$$

Signal

The systematic component that reflects the underlying phenomenon.

Examples: Relationship b/w dosage and recovery, Effect of fertilizer on yield

Noise

Variation that is not completely explained by the variables under study. Sources: Meas. error, Environm. fluctuations, Unobserved variables, Random chance

Why This Matters

Ignoring noise may lead us to discover patterns that do not exist.

Ignoring signal may cause us to overlook important scientific relationships.

Signal versus Noise

$$\text{Observed Data} = \text{Signal} + \text{Noise}$$

Signal

The systematic component that reflects the underlying phenomenon.

Examples: Relationship b/w dosage and recovery, Effect of fertilizer on yield

Noise

Variation that is not completely explained by the variables under study. Sources: Meas. error, Environm. fluctuations, Unobserved variables, Random chance

Why This Matters

Ignoring noise may lead us to discover patterns that do not exist.

Ignoring signal may cause us to overlook important scientific relationships.

Statistical models aim to extract signal while accounting for noise.

Einstein from Noise



Two Sources of Uncertainty

Although uncertainty may arise for many reasons, it is often useful to distinguish between two fundamentally different sources.

Aleatoric Uncertainty

The uncertainty that remains even if we knew the best possible statistical model and all relevant information.

It reflects the inherent variability of the data-generating mechanism.

Epistemic Uncertainty

The uncertainty arising from incomplete knowledge about the data-generating mechanism.

It may often be reduced by collecting more data, measuring additional variables, or improving the statistical model.

Two Sources of Uncertainty

Although uncertainty may arise for many reasons, it is often useful to distinguish between two fundamentally different sources.

Aleatoric Uncertainty

The uncertainty that remains even if we knew the best possible statistical model and all relevant information.

It reflects the inherent variability of the data-generating mechanism.

Epistemic Uncertainty

The uncertainty arising from incomplete knowledge about the data-generating mechanism.

It may often be reduced by collecting more data, measuring additional variables, or improving the statistical model.

Key Observation

Uncertainty usually contains both aleatoric and epistemic components.

Example: Predicting Blood Pressure

Suppose we wish to predict a patient's systolic blood pressure.

Example: Predicting Blood Pressure

Suppose we wish to predict a patient's systolic blood pressure.

$$\text{Blood Pressure} = f(\text{Age}) + \varepsilon$$

Example: Predicting Blood Pressure

Suppose we wish to predict a patient's systolic blood pressure.

$$\text{Blood Pressure} = f(\text{Age}) + \varepsilon$$

The predictions exhibit considerable variability.

Does this mean blood pressure is highly random?

Example: Predicting Blood Pressure

Suppose we wish to predict a patient's systolic blood pressure.

$$\text{Blood Pressure} = f(\text{Age}) + \varepsilon$$

The predictions exhibit considerable variability.

Does this mean blood pressure is highly random?

Not necessarily.

Example: Predicting Blood Pressure

Suppose we wish to predict a patient's systolic blood pressure.

$$\text{Blood Pressure} = f(\text{Age}) + \varepsilon$$

The predictions exhibit considerable variability.

Does this mean blood pressure is highly random?

Not necessarily.

Suppose we build a richer statistical model.

$$\text{Blood Pressure} = f(\text{Age, BMI, Genetics, Medication, Exercise, Diet}) + \varepsilon$$

Example: Predicting Blood Pressure

Suppose we wish to predict a patient's systolic blood pressure.

$$\text{Blood Pressure} = f(\text{Age}) + \varepsilon$$

The predictions exhibit considerable variability.

Does this mean blood pressure is highly random?

Not necessarily.

Suppose we build a richer statistical model.

$$\text{Blood Pressure} = f(\text{Age, BMI, Genetics, Medication, Exercise, Diet}) + \varepsilon$$

The unexplained variability is now substantially smaller.

What Changed?

Did the patient become less random?

What Changed?

Did the patient become less random? **No.** Our understanding improved.

What Changed?

Did the patient become less random? **No.** Our understanding improved.

Source of uncertainty	Interpretation
Previously omitted variables	Epistemic uncertainty (reduced through a better model)
Remaining unexplained variation	Aleatoric uncertainty (inherent variability remaining after modelling)

If you are an omniscient observer then there would be no epistemic error in the system. So, the error decomposition is a property of Nature + Observer + Model.

What Changed?

Did the patient become less random? **No.** Our understanding improved.

Source of uncertainty	Interpretation
Previously omitted variables	Epistemic uncertainty (reduced through a better model)
Remaining unexplained variation	Aleatoric uncertainty (inherent variability remaining after modelling)

If you are an omniscient observer then there would be no epistemic error in the system. So, the error decomposition is a property of Nature + Observer + Model.

Takeaway

Better statistical models often convert what initially appears to be randomness into explainable variation.

Why Does This Distinction Matter?

Suppose a prediction model performs poorly. What should we do?

Why Does This Distinction Matter?

Suppose a prediction model performs poorly. What should we do?

- Collect more data?
- Measure additional variables?
- Build a better model?
- Accept that perfect prediction is impossible?

Why Does This Distinction Matter?

Suppose a prediction model performs poorly. What should we do?

- Collect more data?
- Measure additional variables?
- Build a better model?
- Accept that perfect prediction is impossible?

If uncertainty is

predominantly epistemic

- Collect more data
- Improve the model
- Measure more variables

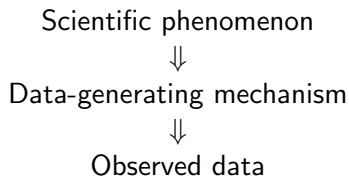
predominantly aleatoric

- Estimate variability
- Quantify uncertainty
- Recognize the limits of prediction

Statistics is not about eliminating uncertainty. It is about understanding, quantifying, and managing uncertainty.

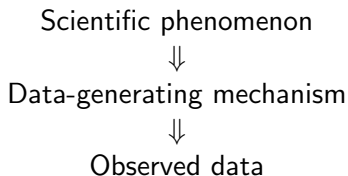
Beyond the Observed Data

Observed data are only the visible outcome. Every dataset is generated by an underlying process.



Beyond the Observed Data

Observed data are only the visible outcome. Every dataset is generated by an underlying process.



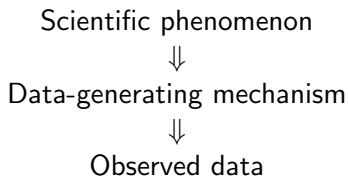
Examples

Human physiology → Blood pressure regulation (age, genetics, diet, medication, stress, ...) → Blood pressure measurements

Consumer behavior → Purchasing decisions (preferences, income, advertising, seasonality, ...) → Customer purchases

Beyond the Observed Data

Observed data are only the visible outcome. Every dataset is generated by an underlying process.



Examples

Human physiology → Blood pressure regulation (age, genetics, diet, medication, stress, ...) → Blood pressure measurements

Consumer behavior → Purchasing decisions (preferences, income, advertising, seasonality, ...) → Customer purchases

Statistics works **backwards**: from the observed data to the hidden mechanism.

The Law of Small Numbers

Two hospitals record births each day.

Hospital A

Approximately 45 births per day.

Hospital B

Approximately 15 births per day.

About 50% of all babies are boys.

On some days, more than 60% of the babies born are boys.

Question

Which hospital is more likely to observe such days?

- Hospital A
- Hospital B
- Both are equally likely

Our Intuition Fails Again

Most people answer

Both hospitals are equally likely

Our Intuition Fails Again

Most people answer

Both hospitals are equally likely

But Randomness Behaves Differently

Smaller samples exhibit much greater variability.

Smaller sample $\implies$ Greater variability

Our Intuition Fails Again

Most people answer

Both hospitals are equally likely

But Randomness Behaves Differently

Smaller samples exhibit much greater variability.

Smaller sample $\implies$ Greater variability

Law of Small Numbers

People tend to believe that small samples should closely resemble the population.

In reality, small samples fluctuate much more than large samples.

Looking Ahead

Today we asked

Why do statisticians think differently?

Looking Ahead

Today we asked

Why do statisticians think differently?

We learned that

→ uncertainty is unavoidable,

Looking Ahead

Today we asked

Why do statisticians think differently?

We learned that

- uncertainty is unavoidable,
- intuition is often misleading,

Looking Ahead

Today we asked

Why do statisticians think differently?

We learned that

- uncertainty is unavoidable,
- intuition is often misleading,
- statistical reasoning relies on probability,

Looking Ahead

Today we asked

Why do statisticians think differently?

We learned that

- uncertainty is unavoidable,
- intuition is often misleading,
- statistical reasoning relies on probability,
- statistical models provide a framework for reasoning under uncertainty.

Looking Ahead

Today we asked

Why do statisticians think differently?

We learned that

- uncertainty is unavoidable,
- intuition is often misleading,
- statistical reasoning relies on probability,
- statistical models provide a framework for reasoning under uncertainty.

Next Lecture

How do probability models represent uncertainty?